

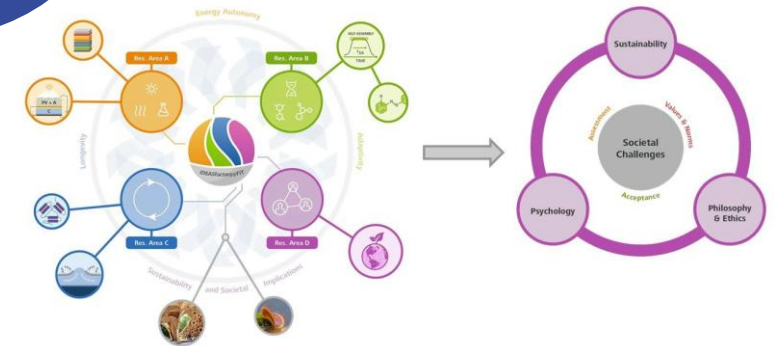
Developing methods for prospective (ethical) technology assessment

- Bachelor in psychology (minor in Philosophy), master in applied statistics
- worked as a "coach for data analysis"
- taught myself web development
- have already some desk rejects, and EU proposal was rejected



PhD
candidate

associated
member



Cluster of Excellence Living, Adaptive and Energy-autonomous
Materials Systems

Empirical Ethics

Necessity of an multi-method approach

Articles:

- Fenn, J., Gorki, M., Höfele, P., Estadieu, L., Becht, C., Kulbe, L., de Wall, M., Gekeler, J., & Kiesel, A. (in prep.). *A Data-driven Multi-Method-Approach for Identifying Ethical Concerns of Emerging Technologies*.
- Fenn, J., Helm, J. F., Höfele, P., Kulbe, L., Ernst, A., & Kiesel, A. (2023). Identifying key-psychological factors influencing the acceptance of yet emerging technologies—A multi-method-approach to inform climate policy. *PLOS Climate*, 2(6), 1–25. <https://doi.org/10.1371/journal.pclm.0000207>
- Fenn, J., Höfele, P., Sölder, P., Reuter, L., & Kiesel, A. (in prep.). *Development and Validation of an Empirical Ethics Scale for Technology Assessment - Challenges and Perspectives for a Real Time Ethics for Emerging Technologies*.

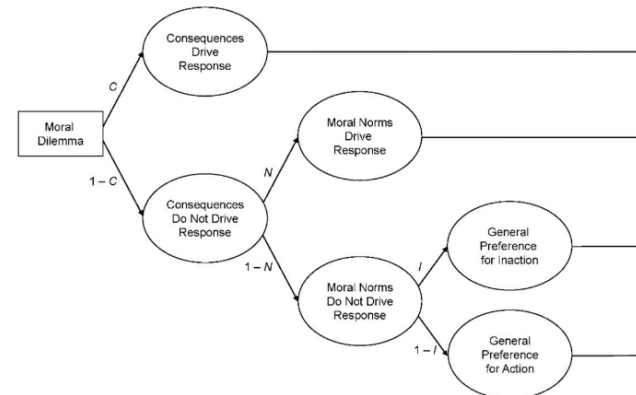
Current “Ethics Scale for Technology Assessment” Team



Paul Sölder (Psychology, Informatics)

- in-depth analysis of Gawronski, B., Armstrong, J., Conway, P., Friesdorf, R., & Hütter, M. (2017). Consequences, norms, and generalized inaction in moral dilemmas: The CNI model of moral decision-making. *Journal of Personality and Social Psychology*, 113, 343–376.

<https://doi.org/10.1037/pspa0000086>



Prof. Andrea Kiesel

Proscriptive Norm Prohibits Action		Proscriptive Norm Prescribes Action	
Benefits of Action Greater than Costs	Benefits of Action Smaller than Costs	Benefits of Action Greater than Costs	Benefits of Action Smaller than Costs
action	inaction	action	inaction
inaction	inaction	action	action
inaction	inaction	inaction	inaction
action	action	action	action



Dr. Philipp Höfele (Philosophy)

<https://www.phil.uni-halle.de/lehrende/hoefe/>

- Practical philosophy and ethics, philosophy of nature and technology
- Board member (secretary) of the "International Schelling Society"



Applied and Empirical Ethics

- Meta ethics, Normative ethics
 - Applied ethics:
 - reflects practical moral dilemmas and considers the “context-sensitivity” of moral theories by embedding them into social practices to formulate recommendations, guidelines or principles
 - Empirical ethics:
 - tries to describe, interpret, explain, evaluate or change social practices by using empirical data
 - Ethical arguments often make use of
 - **Descriptive empirical premises:**
 - “The global average temperature will increase by approximately 2.5 degree Celsius by 2100.”
 - **Normative premises:**
 - “We have the moral obligation to reduce carbon emissions to combat climate change.”
- *integrated empirical ethics* (fundamental interdependence between empirical and normative); *pattern of mixed judgments* (empirical, normative separated) (Düwell, 2009; Molewijk et al., 2004)

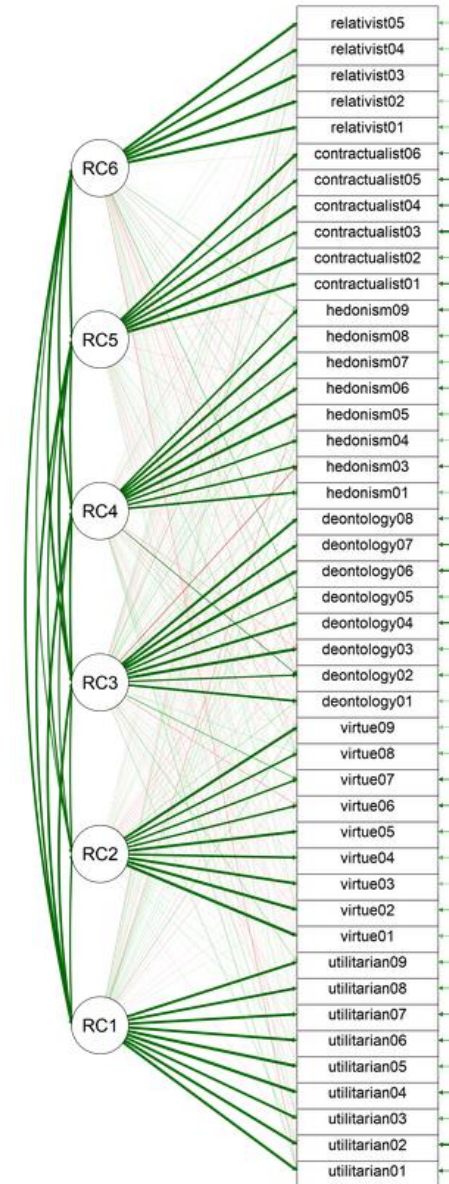
(Paulo & Bublitz, 2020)

“Ethics Scale for Technology Assessment”

Outcome: Survey scale

measures the *degree to which respondents believe a technology is morally acceptable* regarding six **normative ethical theories** encompassing 46 items

- **Utilitarianism:**
 - technology is morally right if it leads to the greatest good for the greatest number
- **Hedonism:**
 - technology is morally right if it increases pleasure, well-being or promotes a good life of an individual who is using the technology



(Jacobs, 2002; Pieper, 2017)

“Ethics Scale for Technology Assessment”

Structure theoretical article, development items

Theoretical part:

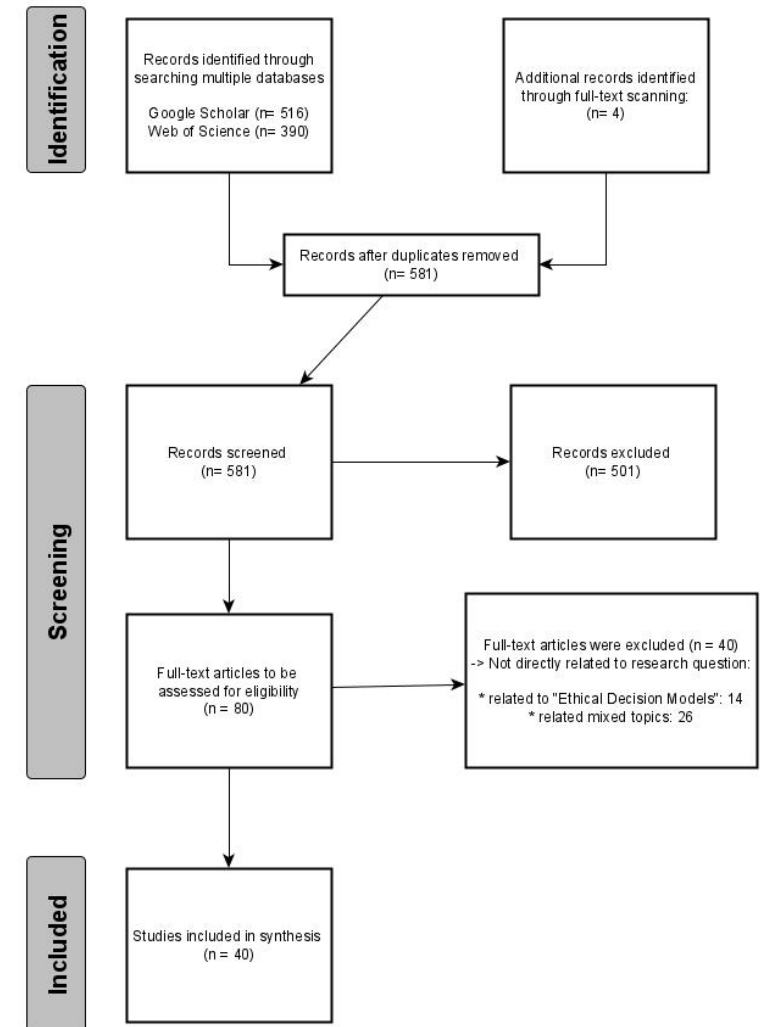
- Central concepts in view of the development of an ethics scale
 - Inter- and transdisciplinary approach
 - Moral psychology and empirical ethics
 - Ethics in real time
 - Ethical technology assessment

Development items:

- Scoping review (existing “ethic scales”), studies that cited the article of Reidenbach and Robin (1990)

Toward the Development of a Multidimensional
Scale for Improving Evaluations of Business Ethics

*R. E. Reidenbach
D. P. Robin*



“Ethics Scale for Technology Assessment”

6 ethic theories, 46 items

Ethics	Description	Items (Examples)
Deontology	A technology is morally acceptable if it respects basic principles and duties. These basic principles are formulated by the personal rationality of morally acting agents and are not conditional on possible consequences of a technology. From an absolutist perspective, these principles should be followed in all circumstances without any partiality or expectations.	# 8 Items • The technology harms (promotes) the autonomy of users. • The technology (does not) go(es) against an important moral rule by which I live.
Utilitarianism	A technology is morally right if it leads to the greatest good for the greatest number. The degree of ethicalness depends on the perceived (possible) outcomes of the technology for all concerned - regardless of whether its “goodness” is equally distributed (criterion of impartiality). The “goodness” can thereby be perceived as utility, the opportunity to decrease harm or the ability of the technology to lead to a better world.	# 9 Items • The technology tends overall to be bad (good) for society. • The technology leads to the least good (greatest good) for the greatest number.
Hedonism	A technology is morally right if it increases pleasure, well-being or promotes a good life of an individual agent, who is using the technology. To increase hedonistic feelings, a technology is used in an instrumental way. In so doing, hedonism overlaps with the teleological theory of egoism.	# 9 Items • The technology is personally unsatisfactory (satisfactory). • The technology minimizes (maximizes) my pleasure.
Virtue ethics	A technology is morally acceptable if it has been developed by someone who has the character and ability to recognize what is morally required. The developer is a genuinely virtuous person and has developed the technology considering her virtuous characteristics.	# 9 Items • The technology is developed by someone who has wrong (good) motivations. • The technology is developed by someone who is acting on ill (good) intentions.
Contractualism	A technology is morally right if it does not violate moral norms that arise from mutual consensus. Not breaking consensus or mutually established contracts is in the interest of moral agents, considering social schemes where these principles were violated. A variety of criteria and conditions can be considered that make a technology morally right (wide reflective equilibrium).	# 6 Items • The technology works against (supports) implicit moral conventions. • The technology violates (does not violate) my ideas of fairness.
Relativism	The moral judgment of a technology depends on the social, cultural and ideological context in which it is implemented, which influences the standards and procedures of justification. In strong relativism, there are only individualized ethical principles. Forms of weak relativism, on the other hand, recognize limited, constrained ethical principles.	# 5 Items • The technology is unacceptable (acceptable) to my culture. • The technology is unacceptable (acceptable) to people I most admire.

“Ethics Scale for Technology Assessment”

Empirical results

Pre-testing scenario texts

Link: https://weblab.psychologie.uni-freiburg.de/publix/5nCCnnL5ykQ?PROLIFIC_PID=111

1. Item Review

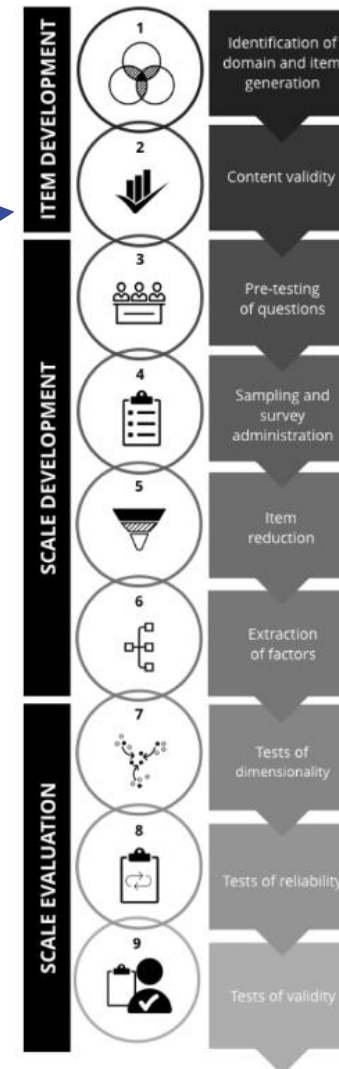
- web probing

2. Examining Dimensionality

- Explorative factor analysis [stakeholders?]

3. Examining the Nomological Net

- Confirmatory factor analysis



From Boateng, G. O., Neilands, T. B., Frongillo, E. A., Melgar-Quinonez, H. R., & Young, S. L. (2018). Best Practices for Developing and Validating Scales for Health, Social, and Behavioral Research: A Primer. *Frontiers in Public Health*, 6, 1–18. <https://doi.org/10.3389/fpubh.2018.00149>; Figure 1

“Ethics Scale for Technology Assessment”

Item Review (Content Validity)

- Link: https://weblab.psychologie.uni-freiburg.de/publix/SBjg8kTUDKo?PROLIFIC_PID=111
 - Category-secelction probe
 - Comprehesion probe

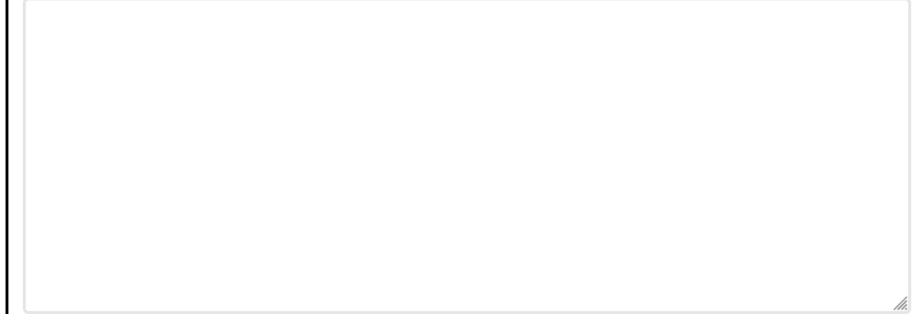
Please indicate on the following scale how you would rate the ethical implications of the technology described in the scenario text. The technology Stratospheric Aerosol Injection...

Read the item and then mark the appropriate answer.

	1	2	3	4	5	6	7	
does not respect an unspoken promise	☐	☐	☐	☐	☐	☐	☐	respects an unspoken promise

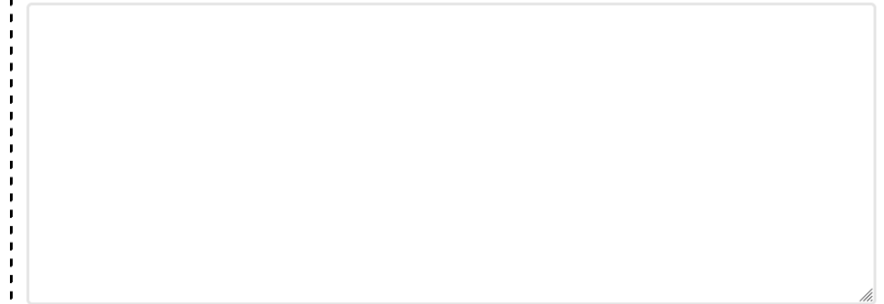
Please explain why you selected "6".

The question was: The technology Stratospheric Aerosol Injection "...does not respect an unspoken promise" (value 1) to "...respects an unspoken promise"(value 7)



What do you consider to be an **unwritten promise** in a technological context?

In doing so, refer to the question: The technology Stratospheric Aerosol Injection "...does not respect an unspoken promise" (value 1) to "...respects an unspoken promise"(value 7), you selected "6"



“Ethics Scale for Technology Assessment”

Item Review (Content Validity): Contractualism

- The technology Stratospheric Aerosol Injection...

"does not respect an unwritten contract"  "respects an unwritten contract"

Contractualist03: Very **few subjects seem to be grasping the idea of an unwritten contract**. Most either state that they have no idea what it would mean, refer to some form of written contract or legislation or just use the opportunity to present their own opinion without even referring to the question, let alone the idea of an unwritten contract. When asked what they consider to be an “unwritten contract”, a substantial amount answers that **they don’t quite know**. The remaining statements are rather diverse and unusually short.

“Ethics Scale for Technology Assessment”

Item Review (Content Validity): Hedonism

- The technology Stratospheric Aerosol Injection...

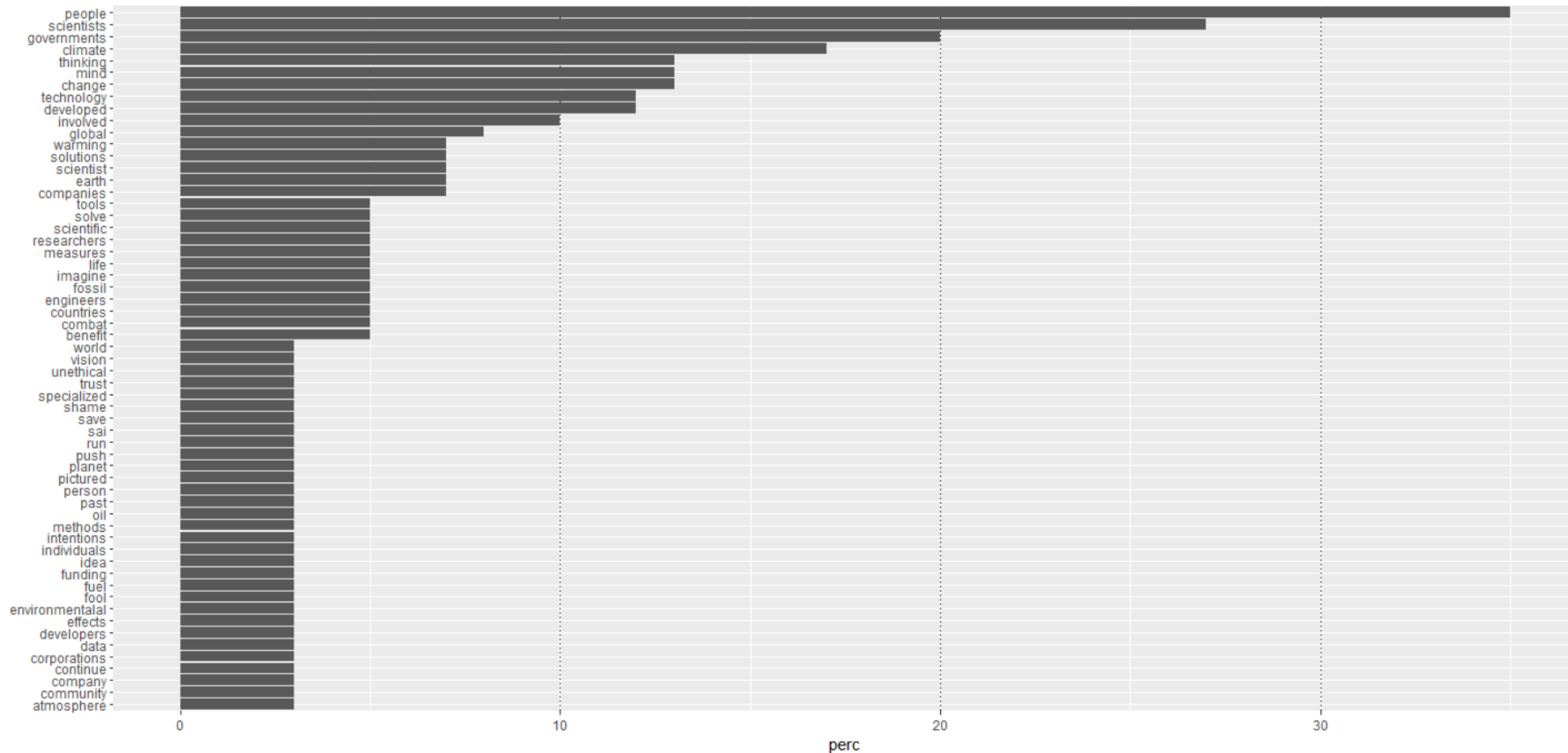
" requires me to make sacrifices in order to use it"  " does not require me to make sacrifices in order to use it"

Hedonism03: Those giving a high value on this item **usually state that SAI doesn't require any sacrifices**, since they personally don't really have to change anything in their behavior. They are thinking about the immediate and personal effect. Making sacrifices means having to change personal behavior to them. Those giving lower values justify it by stating that humanity must make sacrifices or that sacrifices would be needed to stop the root cause of climate change. They are thinking about the broader spectrum and long term consequences. So it seems like **people have very different scopes in mind when answering this question**.

“Ethics Scale for Technology Assessment”

Item Review (Content Validity): Hedonism

- For the development of the technology which people or groups of people did you have in mind?

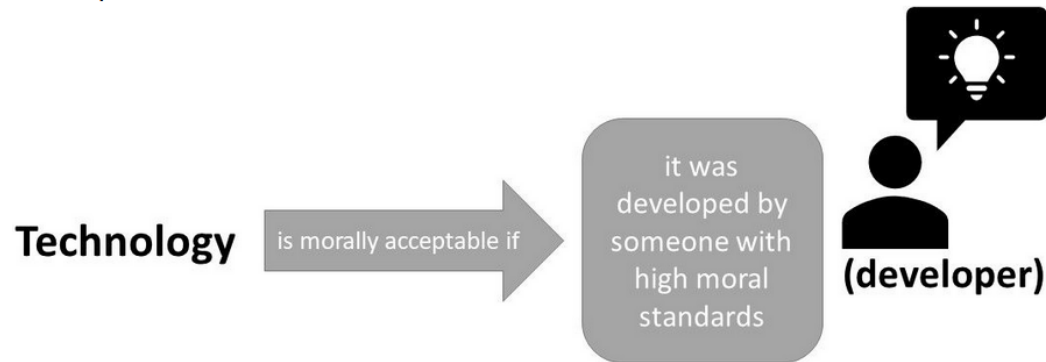


“Ethics Scale for Technology Assessment”

Study Design

Inform participants

Virtue ethics is concerned with the moral standards of the technology's developer:



A technology is morally acceptable if it has been developed by someone who has the character and ability to recognize what is morally required. The developer is an upright person with high moral standards, who has developed the technology on the basis of her/his values.

Subsequently answer scale

Please refer to the explanation of "Virtue ethics" you have just read and indicate on the following scale how you would ethically evaluate the technology. The technology ...

Read the items and then mark the appropriate answers.

	1	2	3	4	5	6	7	
is developed by someone who is indifferent towards nature	☐	☐	☐	☐	☐	☐	☐	is developed by someone who is respectful towards nature
is developed by someone who is insensitive to interactions with society	☐	☐	☐	☐	☐	☐	☐	is developed by someone who is sensitive to interactions with society
is developed by someone who has a bad character	☐	☐	☐	☐	☐	☐	☐	is developed by someone who has a good character
is developed by someone who is not reasonable	☐	☐	☐	☐	☐	☐	☐	is developed by someone who is reasonable

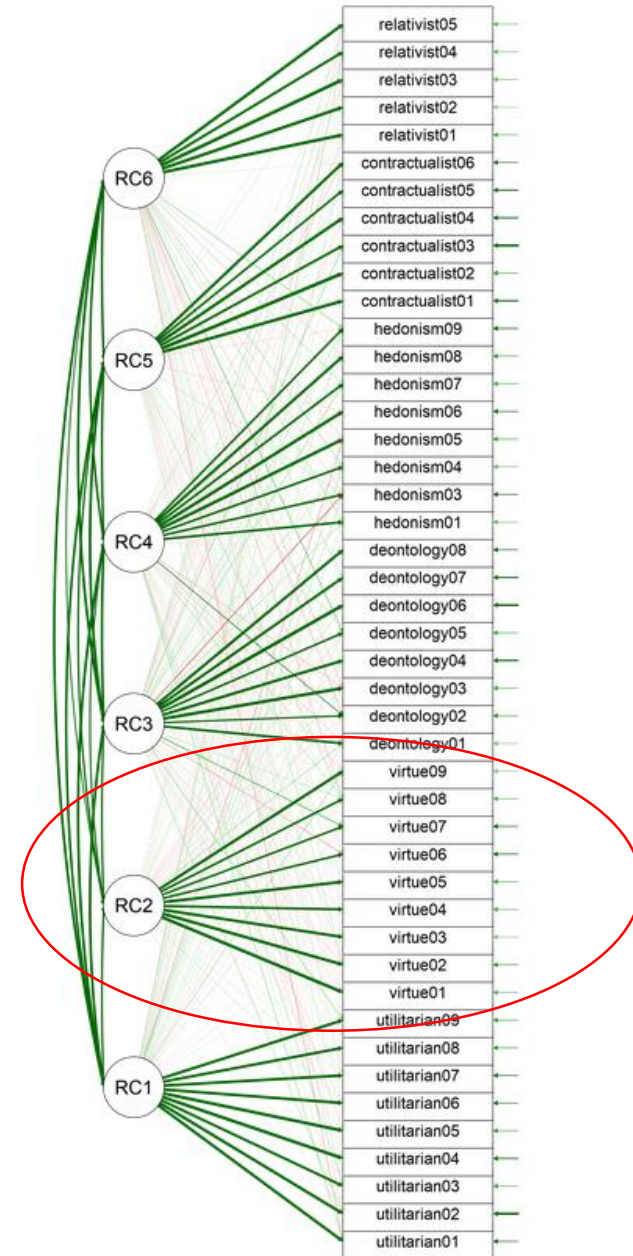
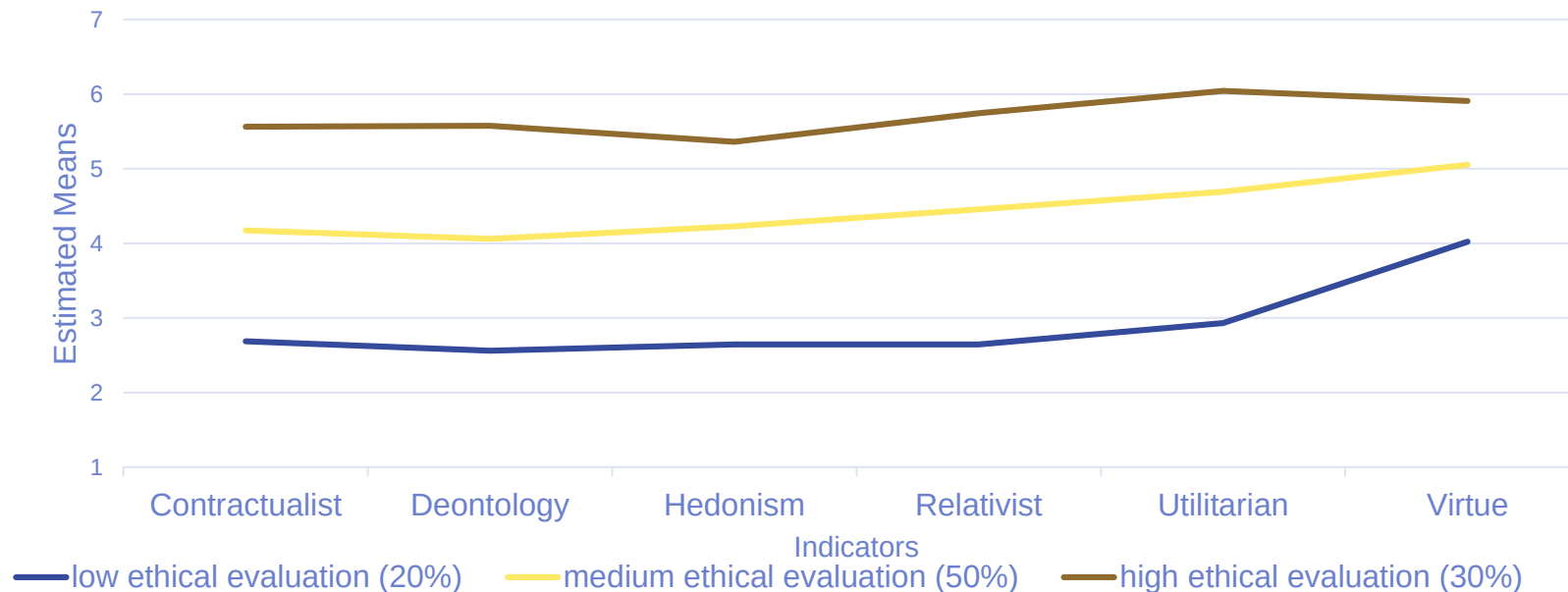
“Ethics Scale for Technology Assessment”

Latent Class Analysis

- but virtue ethics was consistently evaluated higher

Virtue Ethics:

- technology is morally right if it has been developed by someone who has the character and ability to recognize what is morally required



“Ethics Scale for Technology Assessment”

Examining Dimensionality (Test of Dimensionality); Study Design

- Link: https://weblab.psychologie.uni-freiburg.de/publix/I13PgRV83xN?PROLIFIC_PID=111
- 3 technologies (SAI as a narrative, description)

Stratospheric Aerosol Injection (descriptive)



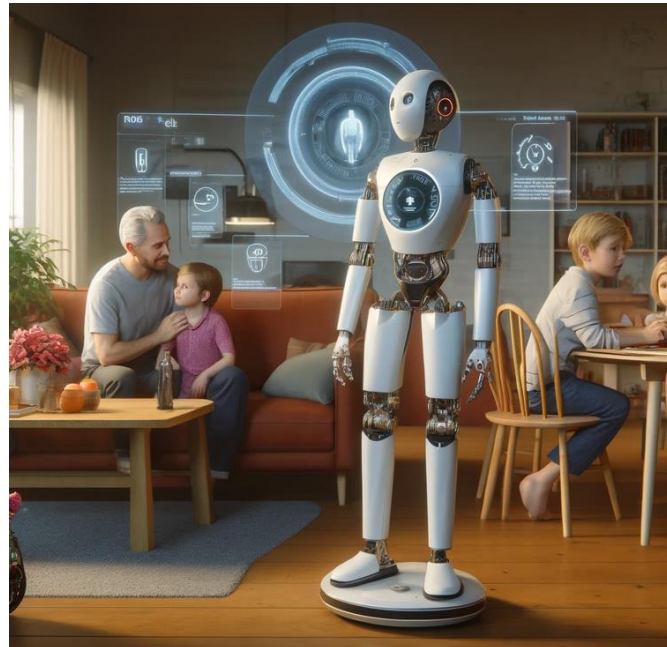
Stratospheric Aerosol Injection (narrative)



Nano-Pat-Parka

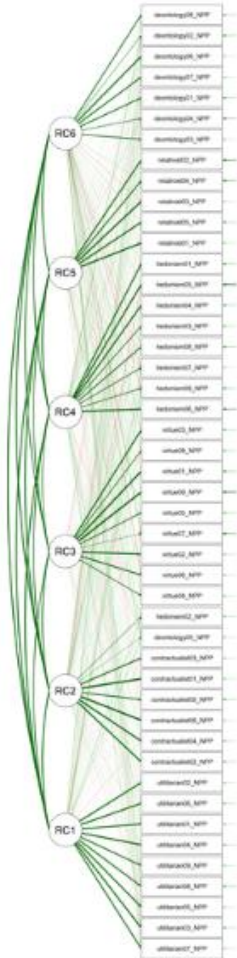


HomeMate



“Ethics Scale for Technology Assessment”

Examining Dimensionality (Test of Dimensionality); define 6 factors



Factor Correlations

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Factor 1	1.000	0.743	0.636	0.768	0.751	0.780
Factor 2	0.743	1.000	0.671	0.653	0.729	0.763
Factor 3	0.636	0.671	1.000	0.560	0.623	0.653
Factor 4	0.768	0.653	0.560	1.000	0.721	0.684
Factor 5	0.751	0.729	0.623	0.721	1.000	0.714
Factor 6	0.780	0.763	0.653	0.684	0.714	1.000



- The technology Nano-Pat-Parka...

“would be selfish for me to use“



“ would not be selfish for me to use “

“Ethics Scale for Technology Assessment”

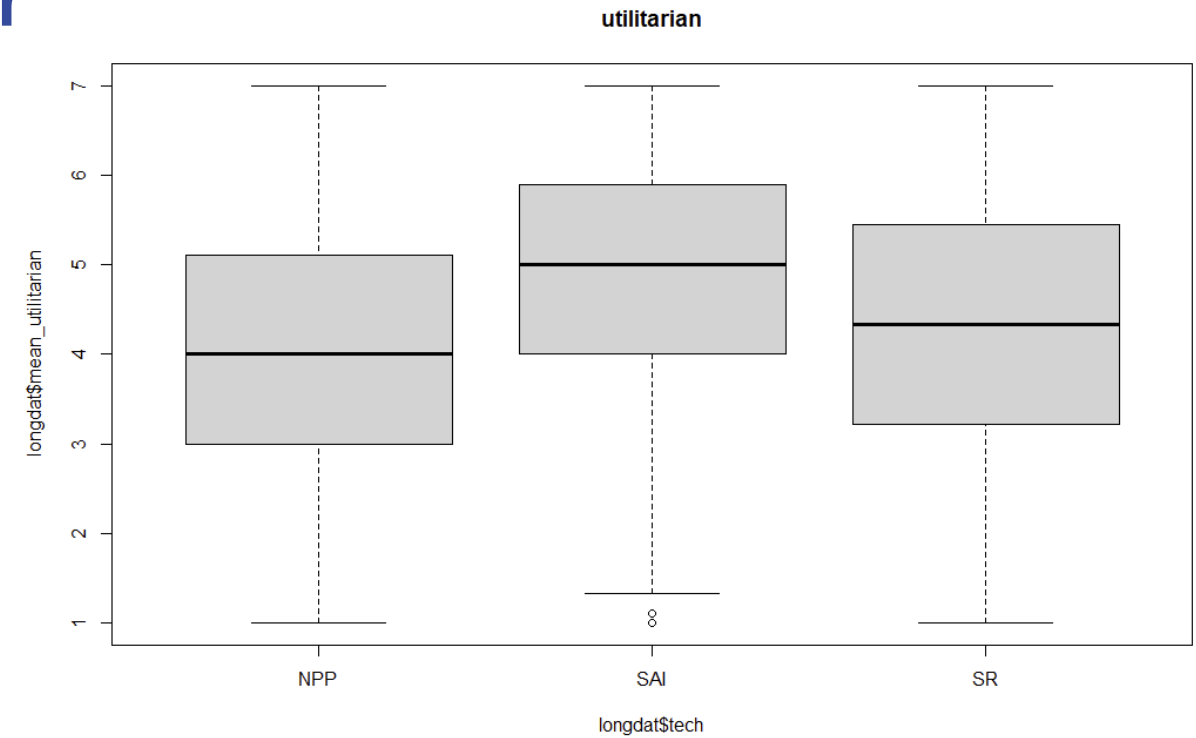
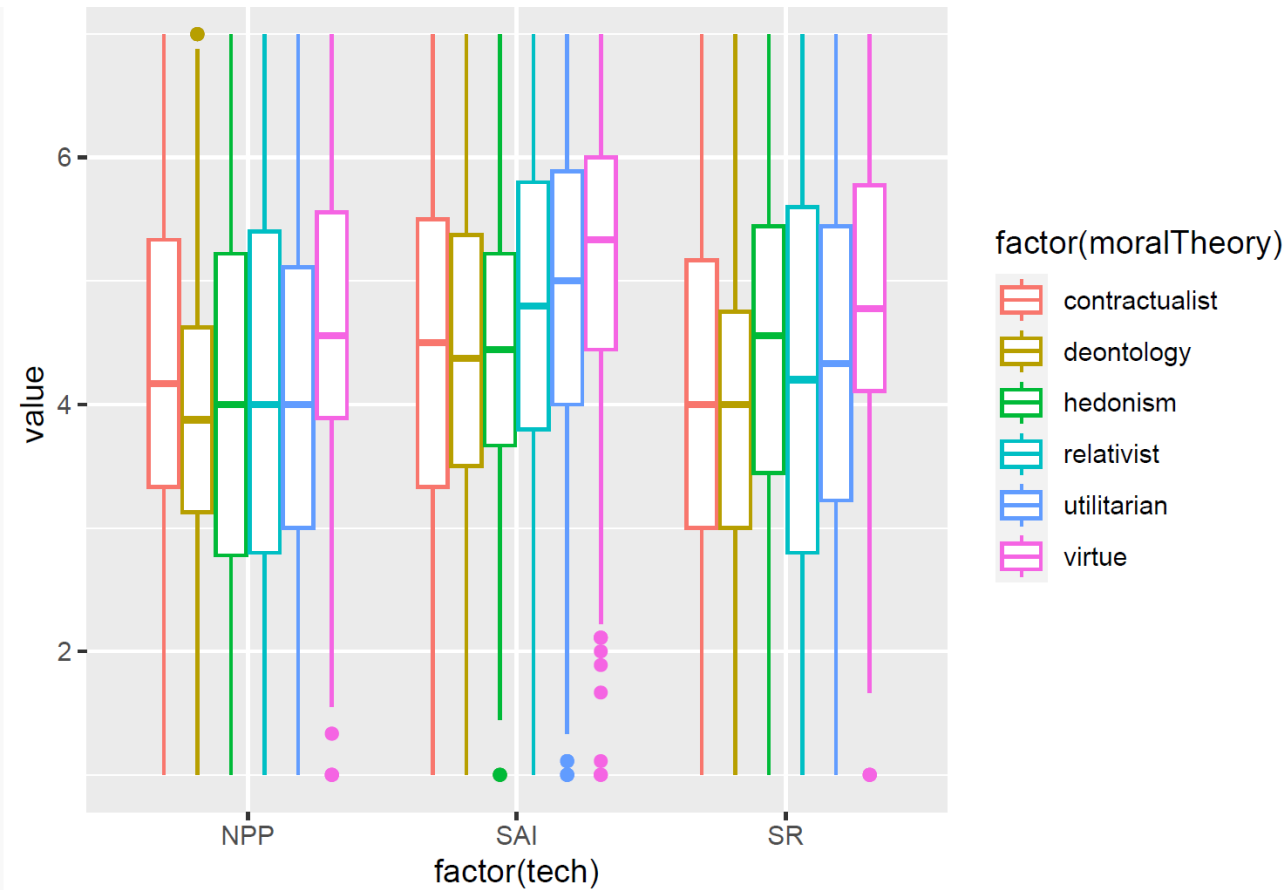
Examining Dimensionality (Test of Dimensionality); parallel analysis

	Factor 1	Factor 2	Factor 3	Uniqueness
hedonism03_SAI		0.575		0.676
hedonism01_SAI		0.809		0.277
hedonism07_SAI		0.874		0.260
hedonism09_SAI		0.888		0.255
hedonism06_SAI		0.878		0.292
hedonism08_SAI		0.677		0.446
hedonism04_SAI		0.844		0.208
hedonism02_SAI		0.417		0.611
hedonism05_SAI		0.675		0.522
relativist04_SAI	0.479	0.463		0.275
relativist03_SAI		0.456		0.410
relativist01_SAI				0.480
relativist05_SAI	0.424	0.429		0.396
relativist02_SAI		0.485		0.353
deontology03_SAI	0.625			0.484
deontology08_SAI	0.589			0.458
deontology06_SAI	0.755			0.349
deontology04_SAI	0.520			0.352
deontology05_SAI				0.416
deontology01_SAI	0.589			0.595
deontology07_SAI	0.761			0.371
deontology02_SAI				0.535
contractualist01_SAI	0.857			0.291
contractualist04_SAI	0.755			0.428
contractualist03_SAI	0.918			0.348
contractualist05_SAI	0.740			0.525
contractualist06_SAI	0.764			0.343
contractualist02_SAI	0.934			0.311
utilitarian04_SAI	0.766			0.210
utilitarian09_SAI	0.784			0.279
utilitarian07_SAI	0.696			0.300
utilitarian06_SAI	0.633			0.328
utilitarian05_SAI	0.721			0.289
utilitarian02_SAI	0.706			0.288
utilitarian01_SAI	0.717			0.294
utilitarian03_SAI	0.646			0.308
utilitarian08_SAI	0.661			0.335
virtue06_SAI			0.642	0.548
virtue07_SAI			0.612	0.329
virtue01_SAI			0.908	0.288
virtue04_SAI			0.582	0.410
virtue05_SAI			0.825	0.248
virtue03_SAI			0.896	0.259
virtue09_SAI			0.923	0.287
virtue08_SAI			0.625	0.405
virtue02_SAI			0.935	0.228

Note. Applied rotation method is promax.

- Hedonism + Relativism
- Deontology + Contractualism + Utilitarianism
- Virtue Ethics

“Ethics Scale for Technology Assessment Examining the Nomological Net



```
> pairs(EMMs, adjust="bon")
contrast estimate SE df t.ratio p.value
SAI - NPP      0.738 0.1076 296  6.865 <.0001
SAI - SR       0.416 0.1114 296  3.731 0.0007
NPP - SR      -0.323 0.0994 296 -3.247 0.0039

P value adjustment: bonferroni method for 3 tests
```

Thanks for your attention and to all people who supported my research

Julius Fenn

Institute of Psychology, Cognition, Action, and
Sustainability

Engelbergerstr. 41, 79085 Freiburg, Germany

julius.fenn@psychologie.uni-freiburg.de

Linktree:



<https://linktr.ee/fennstats>

References

Appendix



Scenario text: Nano-Pat-Parka

Imagine you find yourself in the year 2030. Technological advances in material development and nanotechnology have made it possible for the company NanoTec to develop a new jacket known as the Nano-Pat-Parka.

The text describes the properties and functionalities of the Nano-Pat-Parka:

The bio-inspired Nano-Pat-Parka consists of an inner jacket made of ecological down and a reliable and versatile outer jacket that offers protection from the weather. The outer jacket contains cadmium, a toxic substance. Therefore, the Nano-Pat-Parka is not compostable. The Nano-Pat Parka is adaptive and autonomous. It can adapt to different external and internal conditions with the help of complex sensory systems. This can result in loud, clacking noises and unexpectedly result in a bulky appearance. Due to the smart technology and nanoparticles it contains, the Nano-Pat-Parka is able to learn. Its adaption to external conditions cannot be switched on and off. It is as yet unknown what will become possible with continuous use of the jacket. If there is friction, e.g., when wearing a backpack, the woven molecular N100pX laminate becomes stiff, this can lead to unpleasant smells. In case of contact with stones or objects, the micro-electromechanical inserts in the forearm and back area can dynamically act as protectors. Having a self-illuminating and reflective surface, the outdoor jacket is highly visible and can use solar power to be completely energy-autonomous. The Nano-Pat parka is relatively maintenance-intensive. For example, the built-in technology may start to buzz, which needs to be fixed. By recording and transmitting data (such as the external temperature or body temperature) and GPS data, the company NanoTec aims to improve the Nano-Pat parka continuously. The data sent in this way also allows the Nano-Pat-Parka to call an ambulance completely autonomously in case of an emergency. To this end, the Nano-Pat-Parka is just a prototype but is continuously being tested for its functionality. The main question is whether the properties of the Nano-Pat-Parka are an appropriate response to the changing climatic conditions around the world and the associated need to provide increased protection for people living in challenging environments. This leads to further questions about whether the benefits outweigh the costs for society and what the ethical and societal implications of the Nano-Pat-Parka are.

Scenario text: HomeMate

Imagine you find yourself in the year 2030. Technological advances in social robotics and Artificial Intelligence (AI) have made it possible for the company HomeTec to develop a new social robot known under its brand name HomeMate:

June and Gary Miller were a professionally ambitious couple in their mid-forties. They lived with their three children in a house in an attractive neighborhood on the outskirts of a large city. To make everyday life easier, the Millers decided to order HomeMate. Gary was impressed by HomeMate's initial ability to operate autonomously in the Millers' home. HomeMate understood everyday social situations and responded according to social norms, such as greeting visitors. The latest approaches to robot learning and a sophisticated sensory system enabled HomeMate to learn and adapt its behavior. As a result, HomeMate was soon able to call family members by name and interact with each family member in an appropriate manner. Based on body features such as facial expression or tone of voice, HomeMate was able to recognize the Millers' states of mind and, for example, comforted the children when they cried. Sometimes June and Gary felt guilty because they tended to send HomeMate ahead when the kids were in a bad mood or needed emotional support. HomeMate's human-like design and increasing ability to interact with the Millers led the children to name the robot "Rob". When Rob was playing with the children, it appeared as if they were having real conversations. Only upon closer inspection could one notice that Rob's responses and behavior were quite stereotypical after all. But nevertheless, the children saw the robot more and more as a companion. June noticed with concern that her kids' general behavior was slowly changing. One of her sons even told her that his friends were exhausting and that he would prefer to play with Rob. When the kids were home alone, the parents used Rob's surveillance feature to make sure they weren't doing anything wrong. June had already secretly overheard conversations between her daughter and her first boyfriend. She hadn't been able to resist, but she was actually ashamed of herself. And what if HomeTec, like her, didn't respect privacy and data protection and was surveilling her and her family? One day, the Millers' son Alex reported that one of his classmates also wanted a HomeMate. His parents, however, had told him they did not earn enough money to be able to afford such an expensive robot.

Scenario text: Stratospheric Aerosol Injection (narrative)

Framing Climate Change: According to the latest report by the Intergovernmental Panel on Climate Change (IPCC), the global average surface temperature has risen at a rate of 1°C (1.8°F) compared to 1900. It is extremely likely that this effect is caused by increased emissions of greenhouse gases such as carbon dioxide (CO₂). Greenhouse gas emissions are released for example by burning coal, oil and gas. One degree may not seem like much, but it already has an impact, which can worsen in the future as the temperature continues to rise: Increasingly extreme weather events, like heavy rainfall and storms.

Warming of the oceans due to human influence, which is leading to the melting of the polar ice caps.

Increased acidity of the ocean, because the water absorbs too much of the CO₂ being released, which is leading to the destruction of coral reefs worldwide.

If greenhouse gas emissions continue as they are right now, by 2100 the global temperature could rise up to 3.3 – 5.7°C (5.94 – 10.26°F) compared to before 1900.

SAI Technology: Imagine you find yourself in the year 2030. Despite worrying reports, greenhouse gases have not been sufficiently reduced and governments around the world are beginning to use new technologies to halt global warming.

The scenario describes a typical day in the life of Charlie, a citizen confronted with the new technological developments being used to stop further global warming. Charlie is concerned about the technology, but also sees hope that climate change could be stopped. On a Saturday in the year 2030, Charlie looks up into the sky and sees planes releasing sulfur particles into higher regions of the atmosphere at around 50.000 feet (15 kilometers). Charlie remembers that these sulfur particles reflect sunlight partially back out into space before they warm the earth. He knows that this technique is called "Stratospheric Aerosol Injection", and that it aims to reflect incoming heat from the sun back into outer space to prevent further warming of land and oceans. The cooling effect of this technique was demonstrated by past volcanic eruptions, because, when a volcano erupts, sulfur is also released into the atmosphere. To the best of Charlie's knowledge sulfur is a chemical element found in almost all life forms, and is what gives eggs their distinctive smell for example. Charlie looks at the planes in the sky and thinks about the advantages of "Stratospheric Aerosol Injection" (SAI): SAI is feasible and potentially very effective.

the costs of SAI are low compared with other technologies.

when SAI is deployed, it should lower temperatures within a year.

However, Charlie is very concerned that this technique also comes with certain risks and uncertainties: SAI needs to be deployed for decades or centuries – otherwise temperature could rise even more dramatically within just a few years.

SAI does nothing to counter the inherent cause of climate change – the concentration of carbon dioxide (CO₂) in the atmosphere - and therefore it does not reduce the acidity of the ocean.

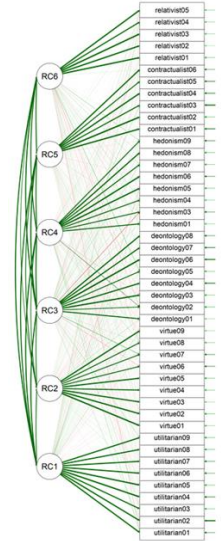
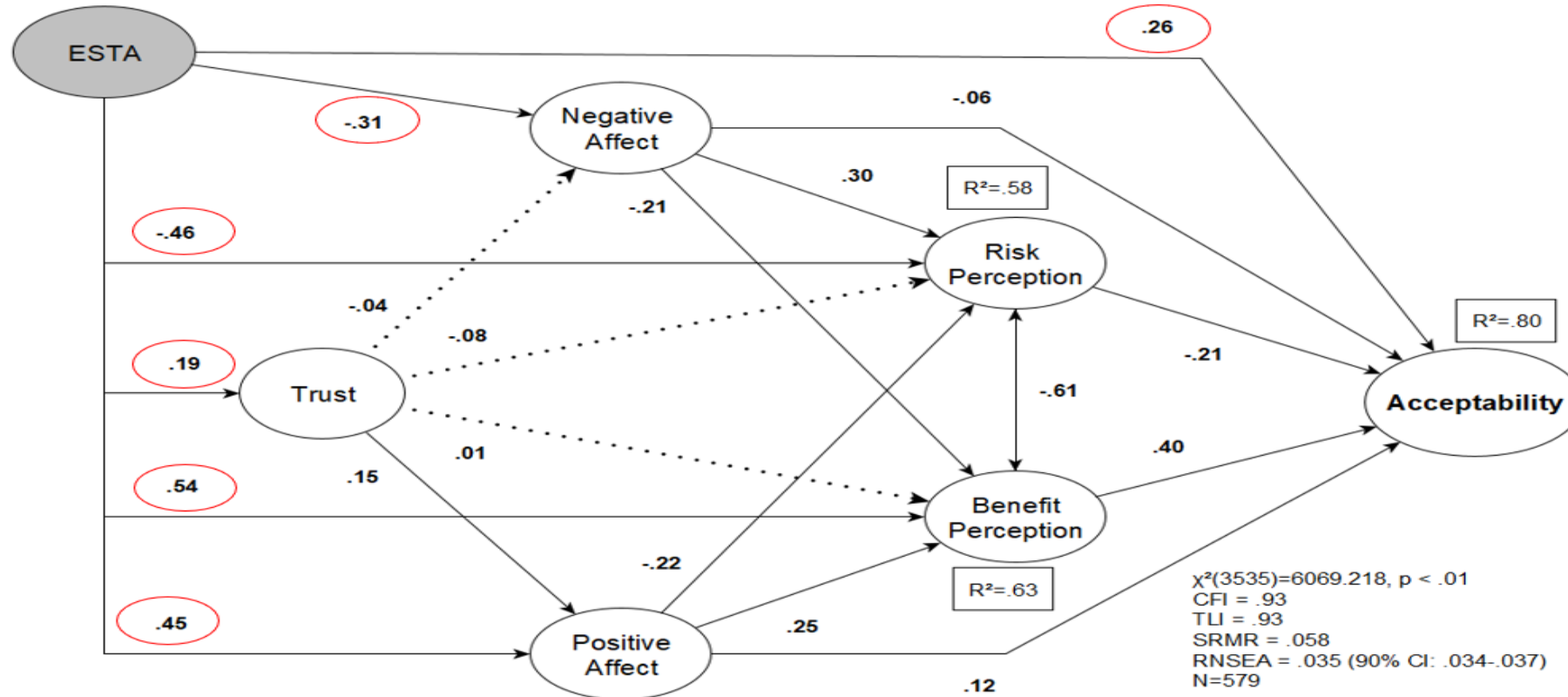
There are already political and social conflicts over its use, as single countries deploying SAI but affecting the whole Earth.

Charlie feels frightened, because potential side effects have not yet been fully resolved and it is unclear what will happen in the future. This technique is being used as an emergency solution, because the temperature has risen beyond the 2°C (3.6°F) target, which nearly all countries in the world had agreed upon. Charlie reflects that, despite SAI, we have to change our lifestyles and transform the economy in a way that limits global warming even further.

“Ethics Scale for Technology Assessment”

predicting central factors of technology assessment

the ethical evaluation of technologies drives core concepts of technological evaluation



The importance of specific ethical arguments

Study design informed by CAMs

Step I procedure



Scenario-based approach

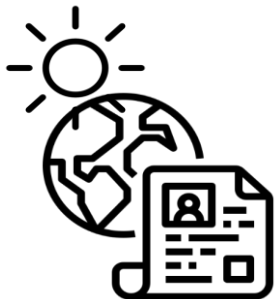
$N = 58$

Cognitive-Affective Maps



informs about potentially missing factors

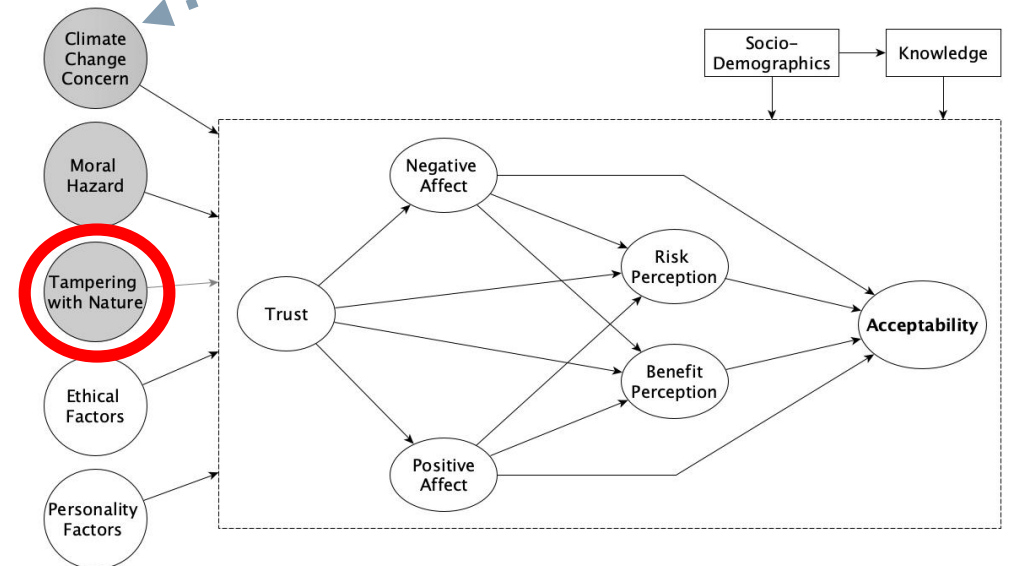
Step II procedure



$N = 579$ (of 600)



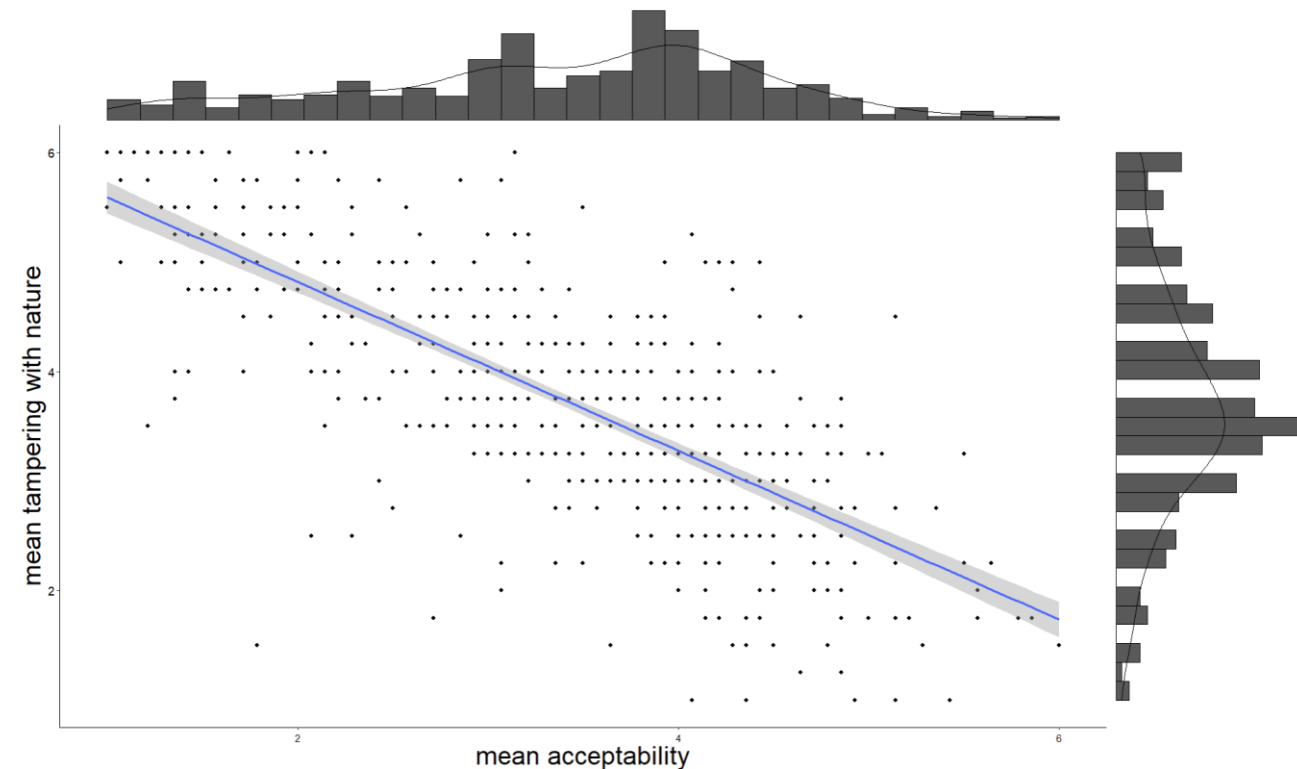
Online Survey



Ethical argument “Do not tamper with nature” existing survey scale

measures the *degree to which respondents believe a technology is disturbing the natural balance in the environment*

Items per scale	Mean	SD	Skewness	Kurtosis (-3)
Tampering with Nature ($\alpha = .89$, $N = 4$)				
SAI is contrary to nature.	4.01	1.19	-0.23	-0.19
SAI technologies disturb the order of nature.	4.22	1.15	-0.36	0.08
Trying to influence the climate system by SAI reflects human arrogance.	3.66	1.37	0.08	-0.77
Human's goal to change the climate system by SAI is immoral.	3.17	1.28	0.32	-0.38



$$r(577) = -.74, CI: [-.78, -.71]$$

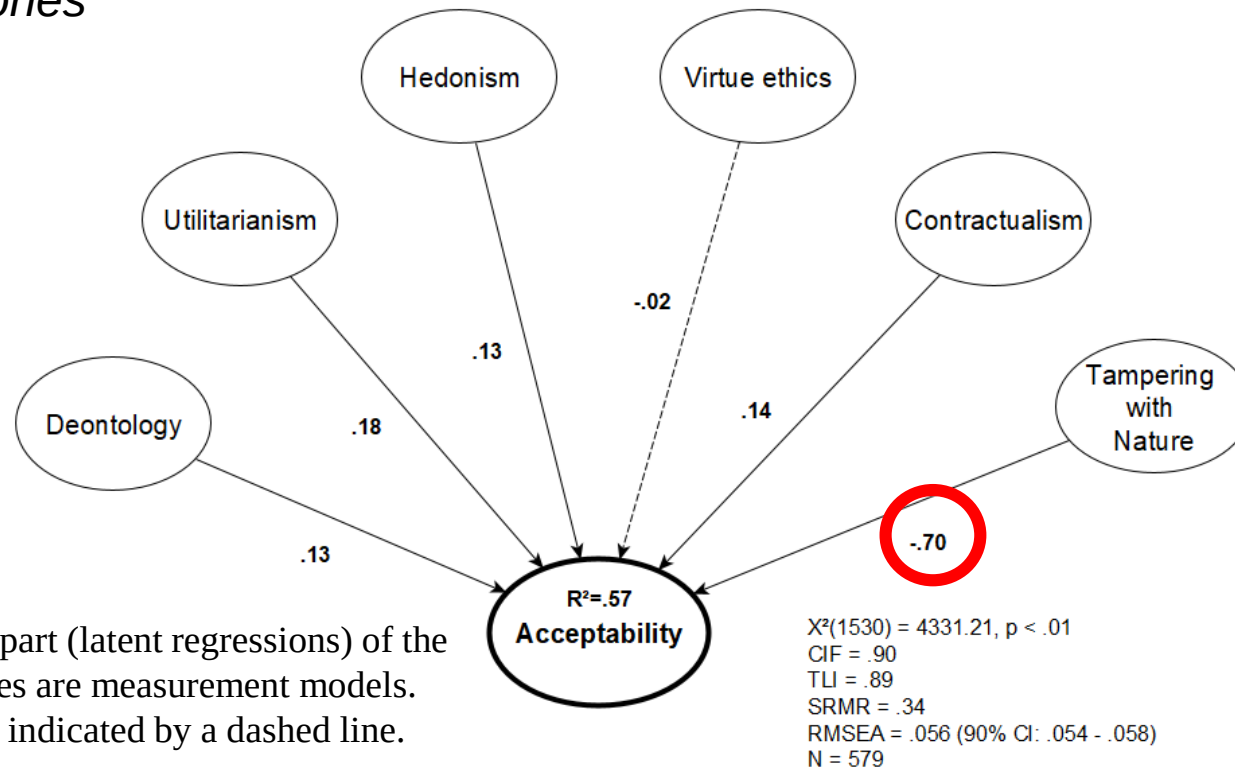
ESTA + ethical argument “Do not tamper with nature” in comparison

ESTA

measures the degree to which respondents believe a technology is morally acceptable regarding six normative ethical theories

“Do not tamper with nature”

measures the degree to which respondents believe a technology is disturbing the natural balance in the environment



Note. Only the structural part (latent regressions) of the model is shown. All circles are measurement models. Non-significant paths are indicated by a dashed line.